

CSCI-665 Foundations of Algorithms

Homework 6

due Tuesday, November 26, 2013

Reading Chapters 25, 26, and parts of 27

Johnson's Algorithm

1. Solve exercise 25.3-1, page 704, using a reduced graph of Figure 25.2 on page 691, namely the graph induced only by 4 vertices $\{1, 2, 4, 5\}$ (as in the previous homework). Show the stages of computation similarly as in Figure 25.6 on page 703.
2. Solve exercise 25.3-4, page 705.
3. Solve exercise 25.3-5, page 705.

Maximum Network Flow Algorithms

4. Interpreting the graph on the next page as a flow network, find the maximum flow from s to t , using the “Edmonds/Karp” algorithm (the second heuristic). At each stage of the algorithm when a new level graph is found you should draw (1) the flow so far and (2) the residual graph and (3) the level graph (draw three separate graphs, not a confusing mass of colors on one graph.) Mark the augmenting paths you find. Draw the final flow.
5. Solve exercise 26.2-2, page 731.
6. Solve exercise 26.2-11, page 731.

